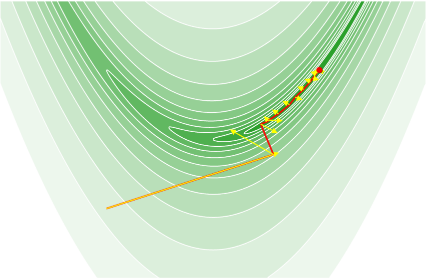
AIE 6001: Advanced Math for Artificial Intelligence

Lecture 0: Introduction



Yang Li

School of AI, CUHKSZ

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Introduction to myself

- Associate Professor, School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen
- Ph.D. in Computer Science, Stanford University, 2017
- Previously worked at Tsinghua Shenzhen International Graduate School
- Research focus: trustworthy machine learning for reliable adaptation, continual learning, interpretable representation, and medical AI.

Note: This course is modified based on AIE6001 course materials developed by Prof. Jie Wang.

Contact and Office Hours

- **Instructor:** Yang Li
- **Email:** `yangl@cuhk.edu.cn`
- **Office:** TX B117

- **Office hour:** Tuesday, 10:30am–12:00pm
- **This week:** no in-person office hour due to conference travel; please feel free to contact me by email.

Teaching Assistants

- Ao Wang: `223010102@link.cuhk.edu.cn`
- Xiaobo Liu: `225080003@link.cuhk.edu.cn`

Course webpage:
`https://yangli-feasibility.com/home/classes/advmath2026fall/`

Course Information (Fall 2026)

- **Course:** AIE6001 – Advanced Math for Artificial Intelligence
- **Units:** 3
- **Medium of instruction:** English
- **Format:** 3 lecture hours per week
- **Prerequisites:** linear algebra, proof-based advanced calculus (or mathematical analysis), calculus-based elementary probability and statistics, and elementary stochastic processes
- **Programming:** experience with Python is helpful, but not compulsory

The course develops mathematical foundations for modern AI, with emphasis on linear algebra, probability, optimization, and information theory.

Who is this course for?

This course is designed to serve the diverse goals of our graduate students.

- **Master's students:** preparing for technically demanding AI/ML industry roles such as ML engineer, data scientist, or quantitative researcher.
- **Ph.D. students:** building the mathematical foundation needed to read, understand, and develop modern AI research.

Why is a strong mathematical foundation critical?

- **For industry:** Technical work in AI relies on a deep understanding of linear algebra, probability, statistics, and optimization. Mathematical depth helps with model design, diagnosis, and adaptation beyond simply using existing libraries.
- **For research:** To innovate, you must first understand. Mathematical fluency is essential for reading and building upon the formal arguments used in modern machine learning papers.

Why Mathematics for AI Practitioners

For Master's Students: Excelling in the Job Market

- **Technical interviews:** A solid grasp of mathematics is often tested directly or indirectly.
  - **Linear algebra:** matrix decompositions (SVD, eigendecomposition), gradients, and their role in neural networks.
  - **Probability & statistics:** distributions, statistical inference, and model evaluation.
  - **Optimization:** gradient descent, convexity, and convergence analysis.
- **Beyond coding:** Understanding why an algorithm works leads to better model design and debugging.
- **Long-term growth:** A strong foundation makes it easier to learn new models and frameworks throughout your career.

Why Mathematics for AI Practitioners

For Ph.D. Students: The Language of Research

- **Reading research papers:** You should be comfortable with
  - proofs of convergence and generalization;
  - statistical bounds and complexity analysis;
  - formal definitions of learning objectives and constraints.
- **Conducting original research:** Mathematics helps you define a problem precisely, analyze its properties, and compare methods rigorously.
- **Publishing:** Clear and correct mathematical reasoning is central to strong technical writing in AI research.

Why I teach this course

- My research in transfer learning, continual learning, and trustworthy AI relies heavily on probability, optimization, information theory, and geometry.
- Mathematical tools are most useful when they help us understand *why* a learning method works, when it may fail, and how it can be improved.
- In this course, I will emphasize both intuition and rigor, and connect the mathematics to modern machine learning models and research papers.
- My goal is to help you build a foundation that remains useful even as particular AI models and software frameworks change.

Course Overview (1)

Linear Algebra for AI

- **Matrix operations:** matrix multiplication, inverses, determinants, and their role in neural networks.
- **Matrix decomposition:** QR factorization, eigenvalue decomposition, SVD, and applications in dimensionality reduction.
- **Vector spaces and norms:** basis, orthogonality, and applications in embeddings.
- **Basics of tensors:** multidimensional arrays for deep learning.

Course Overview (2)

Probability Theory for AI

- **Probability distributions and statistical inference:** common probability distributions and their use in modeling data.
- **Markov chains:** transition matrices, stationary distributions, and applications such as PageRank.
- **Stochastic processes:** discrete- and continuous-time Markov chains and their applications in diffusion models.

Course Overview (3)

Optimization Theory

- **Gradient methods:** stochastic gradient methods and their role in training neural networks.
- **Constrained optimization:** KKT conditions, duality, and applications in SVM.
- **Approximation algorithms:** convex relaxation for non-convex machine learning problems.

Course Overview (4)

Information Theory

- **Entropy and mutual information:** measuring uncertainty and feature relevance.
- **Comparing distributions:** KL divergence, Wasserstein distance, and kernel-based distances.
- **Cross-entropy:** loss functions for classification tasks.

Assessment Scheme

Component	Weight
Assignments	40%
Midterm	25%
Final	30%
Participation	5%
Total	**100%**

Participation

- A short quiz will be released **before each lecture**.
- The quiz typically contains 1–2 multiple-choice questions.
- The quizzes are intended to help you review key concepts and prepare for class.

Teaching Plan (Fall 2026)

Week	Content / Topic
1	General introduction, matrix operations, neural networks
2	Matrix decompositions: QR, eigendecomposition, SVD
3	Vector spaces, norms, tensors, embeddings
4	Probability distributions and statistical inference
5	Markov chains, transition matrices, stationary distributions, PageRank
6	Stochastic processes and diffusion models
7	**Review and Midterm**

Week	Content / Topic
8	Stochastic gradient methods, training neural networks
9	KKT conditions, duality, SVM
10	Convex relaxation for non-convex machine learning problems
11	Information theory, entropy, mutual information
12	KL divergence, Wasserstein distance, kernel-based distances
13	Applications of information theory; review
14	**Final Exam**

Note: The schedule is indicative and may be adjusted based on class progress.

Reading References

- Raman Arora et al., *Theory of Deep Learning*, 2020.
- Wing-Kin Ma, *Matrix Analysis and Computations* (ENGG 5781 lecture notes).
- Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning*.
- George Casella and Roger Berger, *Statistical Inference*.
- Sinho Chewi, *Log-Concave Sampling*.
- James R. Norris, *Markov Chains*.
- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*.
- Stanford EE364b, *Convex Optimization II: Lecture Slides and Notes*.
- Thomas M. Cover and Joy A. Thomas, *Elements of Information Theory*.
- Raymond W. Yeung, *Information Theory and Network Coding*.

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